

Class-Gated Deferral (CGD): Spending LLM Calls Only Where the LLM Is Better

Vasyl Rakivnenko, ...

September 2026

Abstract

We introduce class-gated deferral (CGD), to our knowledge the first cascade algorithm that decides class by class, which model should answer: a computationally ‘free’ (CPU-run) NLP classifier trained on the task’s labels, or a large language model (LLM). For every class the classifier can predict, CGD checks which of the two models is more accurate on held-out data and hands that whole class to the winner, so LLM calls are spent only where the LLM is actually better. Uncertainty-based cascades, such as Online Cascade Learning [5], work differently: they send the LLM whatever the free model is least sure about, even when the LLM is worse on those items. On LegalBench [1], across nine task–LLM combinations, CGD beats both the free classifier and the LLM in seven, and it beats a confidence cascade with the same LLM budget in six, by up to 17 points of balanced accuracy. On CUAD [3] contract clauses with a production API LLM, CGD again beats both models and outperforms a same-budget cascade by 4.8–8.1 points. On Financial PhraseBank [4], it keeps the majority neutral class on the free classifier, cuts LLM calls by 60%, and beats both components in 16 of 28 LLM setups. The gain grows with how class-dependent the LLM’s competence is ($r = 0.57$ across 59 pairs), which suggests a simple test for when to gate by class.

Table 1: LegalBench [1], balanced accuracy (LegalBench’s metric). *Confidence cascade* gives the LLM the same share of items as CGD, escalating the free classifier’s least-confident ones. opp115_data_retention comes from the OPP-115 privacy-policy corpus [2].

Task	LLM	‘Free’ (CPU run) NLP alone	LLM alone	Confidence cascade (same budget)	CGD	Δ vs better	LLM-call share	Outcome
supply_chain_disclosure_best_practice_audits	gpt-oss-120b	74.5%	72.3%	72.2%	82.4% * [†]	+7.9pp	76%	Win (CGD)
supply_chain_disclosure_best_practice_audits	deepseek-v4-pro	74.5%	65.0%	65.7%	82.1% * [†]	+7.7pp	76%	Win (CGD)
supply_chain_disclosure_best_practice_audits	glm-5.3-flash	74.5%	62.9%	63.6%	80.8% * [†]	+6.4pp	76%	Win (CGD)
opp115_data_retention	glm-5.3-flash	79.7%	78.6%	83.1%	81.6% *	+1.9pp	51%	Win (CGD)
opp115_data_retention	kimi-k2.6	79.7%	75.7%	80.8%	81.5% *	+1.8pp	50%	Win (CGD)
opp115_data_retention	gpt-oss-120b	79.7%	73.0%	80.2%	81.4% *	+1.6pp	50%	Win (CGD)
learned_hands_consumer	gpt-oss-120b	84.3%	71.5%	80.2%	85.2% * [†]	+0.9pp	50%	Win (CGD)
learned_hands_consumer	deepseek-v4-pro	84.3%	81.9%	84.8%	84.4%	+0.1pp	42%	Tie
learned_hands_consumer	glm-5.3-flash	84.3%	87.5%	87.5%	87.4%	0.0pp	100%	Tie

Table 2: CUAD [3] contract clauses with a production API LLM. Accuracy (the test sets are class-balanced, so this equals balanced accuracy).

Task	LLM	‘Free’ (CPU run) NLP alone	LLM alone	Confidence cascade (same budget)	CGD	Δ vs better	LLM-call share	Outcome
cuard_covenant_not_to_sue	Jev (TypeSafe API)	94.5%	81.8%	86.9%	95.0% [†]	+0.5pp	50%	Win (CGD)
cuard_change_of_control	Jev (TypeSafe API)	90.6%	82.5%	87.9%	92.7% * [†]	+2.1pp	49%	Win (CGD)

Table 3: Financial PhraseBank [4], accuracy. CGD keeps the neutral class on the free classifier and hands positive and negative items to the LLM (40% of calls). Six of 28 LLM setups shown; CGD beats both components in 16 of the 28.

Task	LLM	‘Free’ (CPU run) NLP alone	LLM alone	Confidence cascade (same budget)	CGD	Δ vs better	LLM-call share	Outcome
financial_phrasebank	grok-4-1-fast-non-reasoning	91.2%	90.0%	94.0%	96.8% *	+5.6pp	40%	Win (CGD)
financial_phrasebank	gpt-5-mini	91.2%	92.8%	97.0%	97.2% *	+4.4pp	40%	Win (CGD)
financial_phrasebank	gpt-5.4-nano	91.2%	92.8%	96.2%	96.8% *	+4.0pp	40%	Win (CGD)
financial_phrasebank	qwen3-1.7b-reasoning	91.2%	78.0%	88.2%	95.2% * [†]	+4.0pp	40%	Win (CGD)
financial_phrasebank	mistral-large-3	91.2%	92.8%	95.0%	95.6% *	+2.8pp	40%	Win (CGD)
financial_phrasebank	gpt-5-nano	91.2%	94.8%	96.8%	97.2% *	+2.4pp	40%	Win (CGD)

‘Free’ (CPU run) NLP alone: a classifier trained on each task’s labeled data (5-fold cross-validation for LegalBench and CUAD; a separate training split for Financial PhraseBank) that runs on CPU. CGD decides which predicted classes go to the LLM from both models’ accuracy on the training folds; all routing decisions are out-of-fold, averaged over 10 fold splits ($\times 3$ cross-validation repeats for LegalBench and CUAD). * $p < 0.05$ vs. the better single model; [†] $p < 0.05$ vs. the same-budget cascade (paired McNemar test for accuracy, paired bootstrap for balanced accuracy). Tie: within 0.5 points.

References

- [1] N. Guha, J. Nyarko, D. E. Ho, C. Ré, et al. “LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models.” arXiv:2308.11462, 2023.
- [2] S. Wilson, F. Schaub, A. A. Dara, F. Liu, S. Cherivirala, P. G. Leon, et al. “The Creation and Analysis of a Website Privacy Policy Corpus.” ACL 2016, pp. 1330–1340.
- [3] D. Hendrycks, C. Burns, A. Chen, S. Ball. “CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review.” NeurIPS 2021 Datasets and Benchmarks Track.
- [4] P. Malo, A. Sinha, P. Korhonen, J. Wallenius, P. Takala. “Good Debt or Bad Debt: Detecting Semantic Orientations in Economic Texts.” Journal of the Association for Information Science and Technology 65(4):782–796, 2014.
- [5] L. Nie, Z. Ding, E. Hu, C. Jermaine, S. Chaudhuri. “Online Cascade Learning for Efficient Inference over Streams.” ICML 2024, PMLR 235:38071–38090.